QuickServe LAN Printing Setup Guide

Prepared by: Stanlly - Software Developer Team Lead

Target: Android Tablet + Termux Print Server

Version: V1.0.4

Overview

QuickServe LAN printing uses a lightweight local print proxy named print-server.js.

The print proxy runs on a device inside the same local network as the thermal printer. QuickServe sends ESC/POS print data to the proxy through HTTP, and the proxy forwards the raw print data to the Ethernet printer over TCP, usually port 9100.

Typical flow:

```
QuickServe Browser
-> HTTP request to print-server.js
-> Android Tablet running Termux
-> TCP port 9100
-> Ethernet thermal printer
```

Prerequisites

- Android tablet connected to the same WiFi/LAN as the printer
- Ethernet thermal printer connected to the LAN by RJ45
- QuickServe print-server.js file
- Printer IP address
- Network access from the Android tablet to the printer

Step 1 - Install Termux And Node.js

Install Termux from F-Droid. Do not use the old Google Play version because it may have outdated packages.

1. Download and install F-Droid.
2. Open F-Droid.
3. Search for Termux.
4. Install Termux.

Open Termux and run:

```
pkg update && pkg upgrade -y
pkg install nodejs -y
termux-setup-storage
```

When Android asks for file permission, allow it. This makes the Downloads folder available from Termux.

Confirm Node.js is installed:

```
node -v
```

Step 2 - Get `print-server.js`

Recommended option from QuickServe:

1. Open QuickServe.
2. Go to Settings -> Printers.
3. Click Proxy Script.
4. Save/download print-server.js.

Other transfer options:

- USB transfer from PC to tablet Downloads
- Google Drive
- Email attachment
- Local file sharing app

Recommended folder:

```
Downloads/print-server.js
```

Step 3 - Start The Print Server

Open Termux and go to Downloads:

```
cd ~/storage/downloads
```

If that folder does not exist, run this once and approve storage permission:

```
termux-setup-storage
```

Then start the proxy:

```
node print-server.js
```

Expected output:

```
QuickServe LAN Print Proxy
Listening on: http://0.0.0.0:3001
Endpoint:    POST /print
Health:      GET  /health
```

The tablet is now acting as the QuickServe print proxy.

Optional custom port:

```
PORT=3002 node print-server.js
```

Step 4 - Find The Tablet IP Address

In Termux, run:

```
ip route get 8.8.8.8
```

Look for the IP after src.

Example:

```
8.8.8.8 via 192.168.1.1 dev wlan0 src 192.168.1.50
```

Tablet IP:

```
192.168.1.50
```

Print Server URL:

```
http://192.168.1.50:3001
```

You can also check:

```
ip -4 addr show wlan0
```

Step 5 - Verify The Print Server

From a browser on the same network, open:

```
http://192.168.1.50:3001/health
```

Expected response:

```
{"status":"ok","server":"print-server","port":3001}
```

If this does not load, check:

- Tablet and QuickServe device are on the same network
- Termux is still running
- Android battery optimization is not stopping Termux

- The IP address is correct

- The port is correct

Step 6 - Configure QuickServe Printer Settings

Open QuickServe and go to:

Settings -> Printers

Then:

1. Click Add Printer.

2. Select Interface: Ethernet.

3. Enter the printer details.

Field	Example
Print Server URL	http://192.168.1.50:3001
Printer IP Address	192.168.1.100
Printer Port	9100

4. Click Test Connection.

5. Confirm the print server is reachable.

6. Assign the printer job, such as Receipt or Kitchen.

7. Save the printer configuration.

Important: Test Connection verifies that QuickServe can reach print-server.js. The final print still depends on the tablet being able to reach the printer IP and port 9100.

Step 7 - Test Printing

After saving the Ethernet printer:

1. Keep Termux running.

2. Complete a test sale or use a print/reprint button in QuickServe.

3. Confirm the printer receives the print job.

If printing fails but /health works, check the printer side:

- Printer IP address

- Printer port, usually 9100

- Printer and tablet are on the same subnet

- Printer is powered on and online

- RJ45 cable/network switch is working

Multi-Printer Setup

QuickServe can save multiple Ethernet printer profiles.

Example:

Printer Name	Printer IP	Common Use
Receipt Printer	192.168.1.102	Payment receipts
Kitchen Printer	192.168.1.100	Kitchen/order list printing
Drink Printer	192.168.1.101	Beverage station

Recommended setup:

1. Add each printer separately.

2. Use the same Print Server URL if all printers are reached through the same Android tablet.

3. Use a different Printer IP Address for each physical printer.

4. Assign jobs/categories in each printer profile.

5. Keep the main receipt printer first if using general receipt/order-list print flows.

Current V1.0.4 note: the printer profiles support jobs and kitchen categories, but some general print paths use the first configured Ethernet printer as the fallback LAN target. Always test receipt, order-list, and kitchen workflows after configuring multiple printers.

Operating Notes

Keep the print server running whenever printing is required.

Recommended:

- Keep Termux open in the background

- Do not close the active Termux session

- Keep the tablet charged

- Disable aggressive battery optimization for Termux if Android stops the process

- Use a stable/static IP for the tablet and printers where possible

Suggested fixed IP setup:

- Reserve the tablet IP in the router DHCP settings

- Reserve each printer IP in the router DHCP settings

- Avoid changing WiFi networks

Troubleshooting

QuickServe Cannot Reach Print Server

Check:

```
http://<tablet-ip>:3001/health
```

If it fails:

- Restart node print-server.js

- Confirm tablet IP

- Confirm QuickServe device and tablet are on the same LAN

- Try the browser health URL from another device on the same WiFi

Print Server Is Reachable But Printer Does Not Print

Check:

- Printer IP address in QuickServe

- Printer port, usually 9100

- Printer is online

- Tablet can reach the printer

- Printer supports raw ESC/POS over TCP

Termux Cannot Find Downloads

Run:

```
termux-setup-storage
cd ~/storage/downloads
ls
```

If needed, use Android file manager to confirm print-server.js is in Downloads.

Server Already Uses Port 3001

Use another port:

```
PORT=3002 node print-server.js
```

Then configure QuickServe with:

```
http://<tablet-ip>:3002
```

Final Workflow

```
User completes sale/order in QuickServe
-> QuickServe builds ESC/POS print data
-> Browser sends data to http://<tablet-ip>:3001/print
-> Termux print-server.js receives the job
-> print-server.js opens TCP connection to <printer-ip>:9100
-> Thermal printer prints automatically
```